

# Proxy Config Security Review — Status Summary

**Revision:** 3 **Last modified:** 2026-07-01T14:48:00Z **Status:**  
Companion summary of [Status.md](#) (§11.4.56 two-audience).

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## Page 1 — For the operator / stakeholders (plain language)

We did a careful read-through of the proxy's configuration — both the web proxy (Squid) and the SOCKS5 proxy (Dante) — looking for anything that could **leak information about you or your network**, or that could be **abused if the proxy were ever exposed to the open internet**. Because this is a VPN proxy whose whole job is to hide who and where you are, any such leak matters.

### The good news — already solid:

- **Access rules are locked down.** The web proxy only allows secure HTTPS connections on the standard port, denies everything else by default, and keeps its admin interface local-only.
- **No HTTPS "man-in-the-middle" surface.** The proxy passes your encrypted traffic straight through instead of opening it up, so there is nothing there to get wrong.

### Being fixed right now (small leaks, no downside to fixing):

- **The proxy was announcing itself.** It added a header revealing its internal name and exact software version to every site you visit — we confirmed this live and are turning it off.
- **Error pages showed the software version** — being suppressed.
- **Your local IP address could be forwarded** on non-encrypted requests. This is the most privacy-sensitive of the four. We are turning it off as a safety measure. (Honest note: we inferred this from the software's default behaviour but could not reproduce it live, so we are fixing it defensively rather than claiming we saw it happen.)

- **The container's internal id could leak** on error pages — being replaced with a neutral name.

**Needs your decision (we did NOT change these on our own, because the tight fix could break your connection):**

- **DNS could leak** in one operating mode (the web proxy asks a public DNS server directly instead of the private encrypted one). The other mode is already protected. Pointing it at the private resolver is the fix — but only if that resolver is reachable, which is your call.
- **The SOCKS5 proxy would accept anyone** if its port ever escaped the private network, and it could be pointed at internal/cloud-metadata addresses. We applied the safe part of the fix now; locking it down fully needs you to confirm the trusted client range and internal targets so we don't cut off legitimate traffic.

**Bottom line:** the core access controls are already strong; four minor information leaks are being closed this round with zero connectivity risk; three larger items are flagged for your decision because the safest-possible fix could otherwise break connectivity. One leak was confirmed live; one was fixed defensively and honestly labelled as inferred, not confirmed.

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## Page 2 — For software engineers

Static config-security review of the proxy data plane (Squid non-intercepting forward proxy + Dante SOCKS5). Findings are FACTs from the review (§11.4.6). Status: **HARDENED / FIXED** (RED→GREEN proven this round) / **TRACKED** (lands this round) / **TRACKED** (operator decision, §11.4.101).

| # | Area              | Finding / directive                                                                                                         | Sev | Status         | Evidence                                        |
|---|-------------------|-----------------------------------------------------------------------------------------------------------------------------|-----|----------------|-------------------------------------------------|
| 1 | Squid ACL chain   | deny CONNECT !SSL_ports (443-only) · http_access deny all terminator · manager localhost-only · fail-closed dynamic profile | —   | HARDENED       | Config (static)                                 |
| 2 | Squid TLS         | CONNECT pass-through; no https_port/ssl_bump — no intercept surface by design                                               | —   | HARDENED (N/A) | By-design                                       |
| 3 | Squid via         | via on → Via: 1.1 proxy-squid (squid/6.13) (hostname + version) → <b>via off</b>                                            | MED | FIXED          | CONFIRMED<br>qa-regression<br>headers<br>red_20 |
| 4 | Squid version str | httpd_suppress_version_string off → version on ERR pages                                                                    | MED | FIXED          | Squid disclosure                                |

| # | Area                      | Finding / directive                                                                                                            | Sev  | Status                | Evidence                                                                                                                                                                                                                                                                                                         |
|---|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                           | incl. ERR_TUNNEL_DOWN →<br><b>httpd_suppress_version_string on</b>                                                             |      |                       |                                                                                                                                                                                                                                                                                                                  |
| 5 | Squid<br>forwarded_for    | default → X-Forwarded-For:<br>client-LAN-IP on plain HTTP<br>(privacy-defeating) →<br><b>forwarded_for delete</b>              | HIGH | FIXED                 | <b>INFERRED</b><br>default<br>reproducible<br>endpoint<br>defensive<br>(§11.4.101)                                                                                                                                                                                                                               |
| 6 | Squid<br>visible_hostname | absent → container-id leak →<br><b>visible_hostname helix-proxy</b>                                                            | LOW  | FIXED                 | Squid<br>fallback                                                                                                                                                                                                                                                                                                |
| 7 | Squid<br>dns_nameservers  | dns_nameservers 8.8.8.8<br>bypasses DoT dnsproxy →<br>DNS leak in <b>static</b> mode<br>(dynamic mitigated by<br>never_direct) | MED  | TRACKED<br>(operator) | Re-producible<br>loopback<br>risk (§11.4.101)                                                                                                                                                                                                                                                                    |
| 8 | Dante auth/<br>client     | socksmethod none +<br>client pass from:0.0.0.0/0 →<br>open relay if :51080 escapes<br>the bridge                               | HIGH | TRACKED               | Safe<br>terminating<br>client<br>= connection<br>sockd.conf(5)<br>command<br>socks<br>169.254.169.254<br>172.17.0.1<br>— REPRODUCIBLE<br>internal<br>fast, client<br>external<br>PASS<br>security<br>verdict<br>(§11.4.101)<br>require<br>authentic<br>block<br>(§11.4.101)<br>time;<br>standards<br>(§11.4.101) |
| 9 | Dante egress              | socks pass to:0.0.0.0/0, no<br>command: → SSRF to<br>169.254.169.254/RFC1918 +<br>BIND                                         | HIGH | FIXED                 |                                                                                                                                                                                                                                                                                                                  |

**References:** squid-cache.org config docs (via, httpd\_suppress\_version\_string, forwarded\_for, visible\_hostname, dns\_nameservers, never\_direct); Dante sockd.conf(5); OWASP Server-Side Request Forgery (SSRF).

Composes §11.4.45 (integration-status doc), §11.4.56 (two-audience summary), §11.4.6 (no-guessing — CONFIRMED via vs INFERRED X-Forwarded-For distinguished), §11.4.101

(autonomous-decision-over-blocking — connectivity-risky fixes  
TRACKED for operator), §11.4.5/§11.4.69 (captured evidence for  
the confirmed leak).